

SmolVLM

Redefining Small and Efficient Multimodal Models

Hugging Face & Stanford University | [arXiv:2504.05299](#)

Compact vision-language models achieving competitive performance with sub-1GB GPU RAM

SmolVLM-256M

0.8 GB

SmolVLM-500M

1.2 GB

SmolVLM-2.2B

4.9 GB

Large VLMs Are Powerful but Impractical for Edge Deployment

State-of-the-art VLMs require massive GPU memory and compute

Molmo-A1B-7B needs 27.7 GB RAM — exceeds most edge devices

Idefics-80B: 80B parameters, 300× larger than SmolVLM-256M

Parameter count does not equal efficiency

Goal: competitive performance at sub-1GB RAM

Molmo-A1B-7B

27.7 GB RAM • 7B parameters

SmolVLM-256M

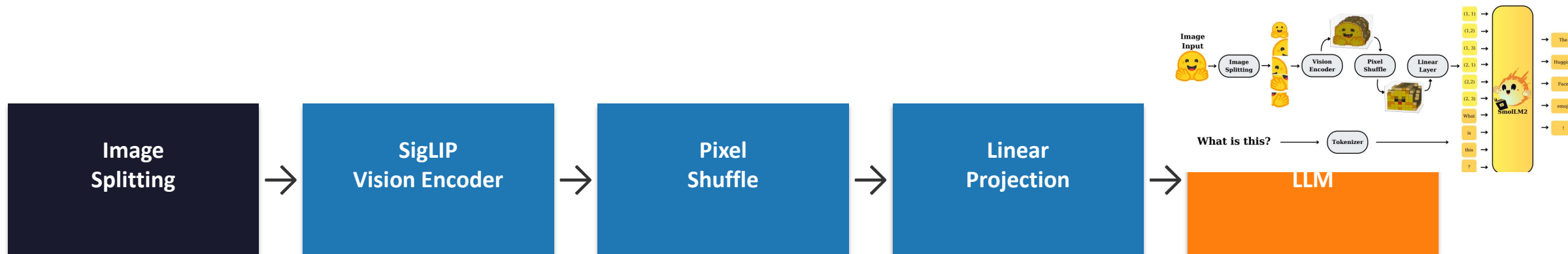
0.8 GB RAM • 256M parameters

SmolVLM-2.2B

4.9 GB RAM • 2.2B parameters

SmolVLM-256M outperforms the 300× larger Idefics-80B

SmolVLM Architecture: Image Splitting, Pixel Shuffle, and Smol



Images split into sub-images → encoded by SigLIP vision encoder

Pixel shuffle (space-to-depth) compresses visual tokens by r^2

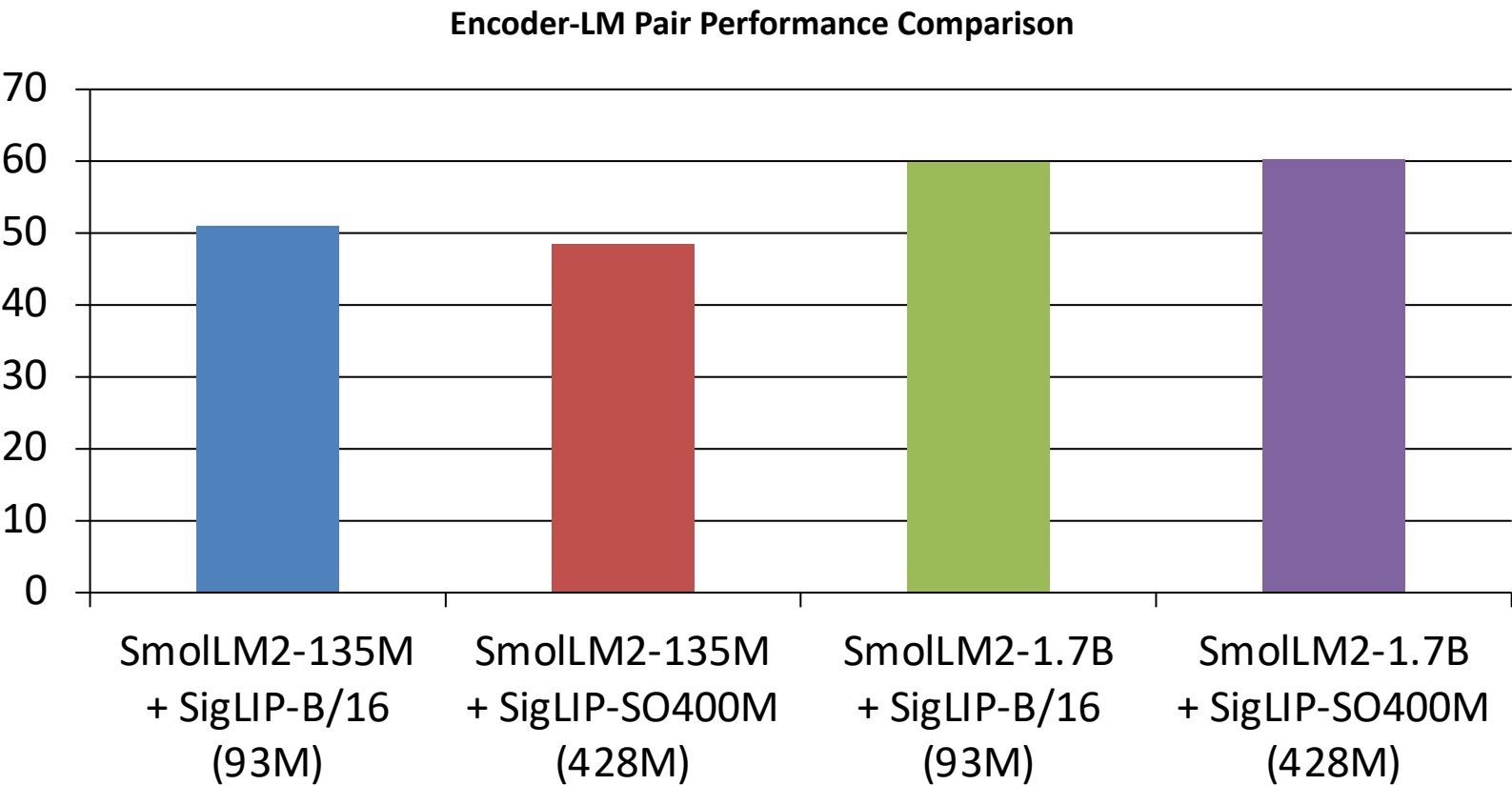
Linear projection maps to SmolLM2 input space

Text tokens concatenated with visual tokens → autoregressive generation

Video: frames sampled and processed with the same pipeline

Smaller Vision Encoders Outperform Larger Ones in Compact V

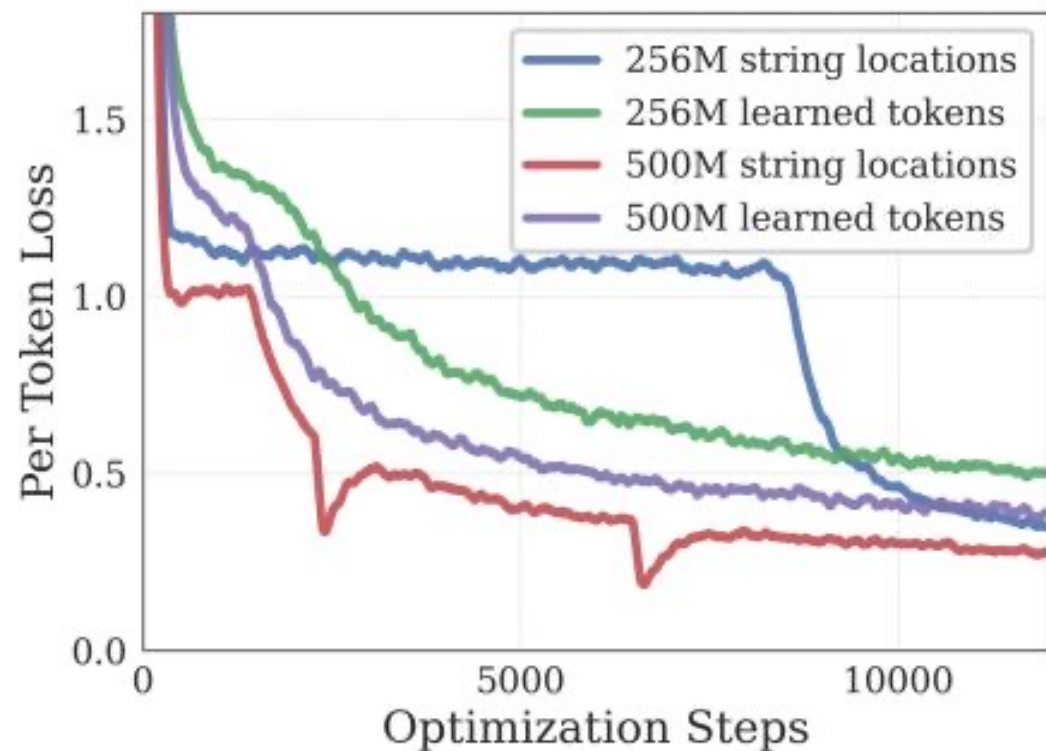
Finding 1: Balanced encoder-LM parameter allocation matters



critical for efficiency.

Aggressive Pixel Shuffle (r=4) Cuts Visual Tokens 16× for Small

Finding 3: Token compression via space-to-depth rearrangement



fine-grained visual details.

Extending Context from 2K to 16K Tokens Unlocks Higher Image

Finding 2: Longer context enables finer-grained visual understanding



RoPE base increased from 10K to 273K to support extended context

Fine-tuned on mixed long/short-context data for robust generalization

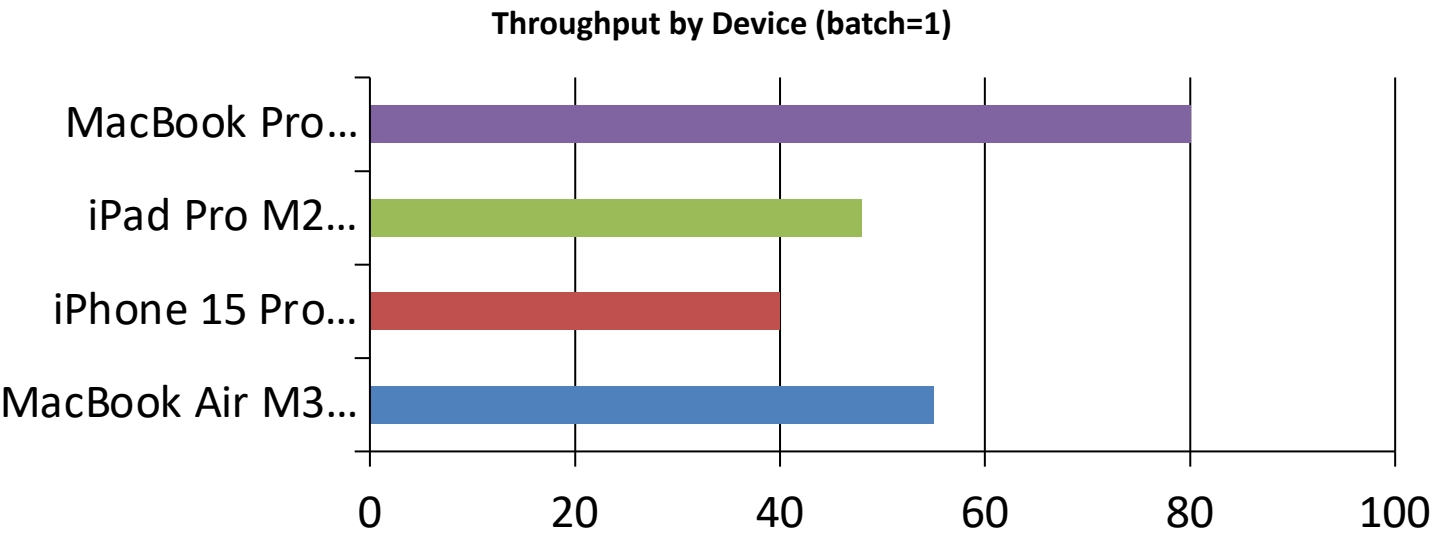
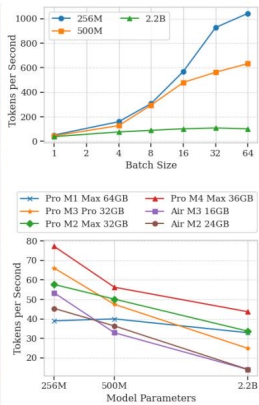
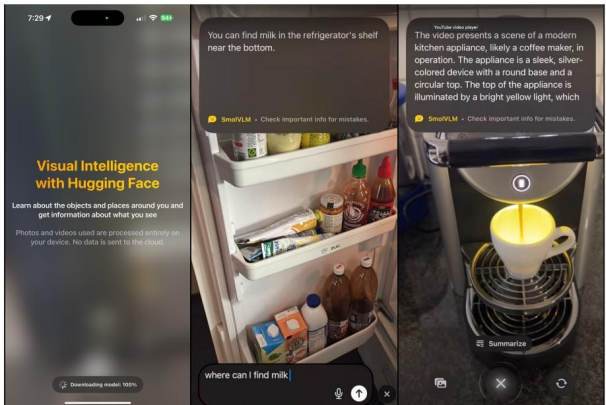
Longer context allows higher image resolution inputs without OOM

2.2B model uses 16K context; smaller variants use 8K

SmolVLM Achieves Competitive Performance at Sub-1GB to 4.9GB

Benchmark	SmolVLM 256M	SmolVLM 500M	SmolVLM 2.2B	Best Efficient OS	RAM (GB)
OCRBench	52.6%	61.0%	72.9%	Molmo-A1B-7B	0.8–4.9
AI2D	46.4%	59.2%	70.0%	Molmo-A1B-7B	—
ChartQA	55.6%	62.8%	68.7%	Molmo-A1B-7B	—
TextVQA	50.2%	60.2%	73.0%	Molmo-A1B-7B	—
DocVQA	58.3%	70.5%	80.0%	Molmo-A1B-7B	—
ScienceQA	73.8%	80.0%	89.6%	Molmo-A1B-7B	—
MMMU	29.0%	33.7%	42.0%	Molmo-A1B-7B	—
MathVista	35.9%	40.1%	51.5%	Molmo-A1B-7B	—
MMStar	34.6%	38.3%	46.0%	Molmo-A1B-7B	—
Video-MME	33.7%	42.2%	52.1%	InternVL2-2B	—
MLVU	40.6%	47.3%	55.2%	InternVL2-2B	—
AVG (All)	44.0%	51.0%	59.8%	—	—
RAM Usage: SmolVLM-256M = 0.8 GB SmolVLM-500M = 1.2 GB SmolVLM-2.2B = 4.9 GB Molmo-A1B-7B = 27.7 GB					

Real-Time Inference on iPhones: Up to 80 Tokens/Second on A



256M model

~40–80 tokens/s

256M + batch=64

~1000 tokens/s

Fully on-device

No cloud dependency

HuggingSnap iOS app: processes photos and videos entirely on-device via Hugging Face Transformers. Throughput scales roughly linearly with batch size for the smallest model.

Key Takeaways: Design Principles for Efficient Multimodal Mo

1

Balance Encoder-LM Compute

A smaller vision encoder (93M) outperforms a larger one (428M) when paired with small language models. Only at 1.7B+ LM scale does the larger encoder justify its overhead.

2

Extend Context Length

RoPE base adjustment from 10K to 273K + mixed-context fine-tuning enables 16K context, allowing higher resolution inputs without OOM.

3

Use Aggressive Token Compression for Small Models

Pixel shuffle at $r=4$ delivers 16× token reduction, easing attention and improving long-context modeling for compact VLMs.

4

Sub-1GB Inference Is Achievable

SmolVLM-256M runs in 0.8 GB GPU RAM, enabling deployment on smartphones and low-power edge devices.

5

Fully Open-Source

All weights, datasets, code, and the HuggingSnap mobile app are publicly released to promote reproducibility.